

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, SPRING SEMESTER 2023–2024

MATHEMATICS FOR COMPUTER SCIENTISTS 2

Time allowed: TWO Hours

This is a mock exam to help prepare for the ExamSys exam.

Answer ALL FOUR QUESTIONS

This examination will be marked out of 100.

Calculators from List A are permitted in this examination as well as one two-sided A4 sheet of notes, but otherwise this is a closed-book examination, and students may not refer to any other source or person in taking this paper.

List A

Basic models	Scientific Calculators
Aurora HC133	Aurora AX-582
Casio HS-5D	Casio FX82 family
Deli – DL1654	Casio FX83 family
Sharp EL-233	Casio FX85 family
	Casio FX350 family
	Casio FX570 family
	Casio FX991 family
	Sharp EL-531 family
	Texas Instruments TI-30 family
	Texas BA II+ family

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted. You are not permitted to take any paper or notes out of the examination room during or after the examination. Any rough notes that you make on the paper provided will be collected by staff and destroyed.

1. Systems of linear equations

(a) Consider the following system of linear equations

$$\begin{cases} z - 4w = -4 \\ y - 3z + 14 = 13 \\ 3x + z + w = 1 \\ x + w = 2. \end{cases}$$

Determine whether or not the system is consistent, and if it is, give a particular solution.

[10 Marks]

(b) Compute the reduced row echelon form of the matrix

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 1 \\ 1 & 3 & -2 & 0 & -1 \\ 2 & 6 & -3 & 1 & 0 \end{pmatrix}.$$

[5 Marks]

Suppose a system of linear equations is represented as an augmented matrix with reduced row echelon form

$$\left(\begin{array}{ccc|c} 1 & 0 & 4 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

(c) Is the system consistent?

[5 Marks]

(d) Give a basis for the solution set of the corresponding homogeneous system.

[5 Marks]

[End of Question 1: Total 25 marks]

Solution

(a) The system is consistent with (unique) solution $x = 5/2$, $y = -19$, $z = -6$, $w = -1/2$.

(b) The reduced row echelon form is

$$\begin{pmatrix} 1 & 3 & 0 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

(c) The system is inconsistent, since there's a pivot in the augmentation column.

(d) A basis for the solution set of the homogeneous system is:

$$\left(\begin{pmatrix} -4 \\ -1 \\ 1 \end{pmatrix} \right)$$

2. Linear maps & basis

Suppose that $\mathcal{B} = (u_1, u_2, u_3, u_4)$ is a basis for $\mathbf{R}^4$ and $\mathcal{C} = (v_1, v_2, v_3, v_4)$ is a basis for $\mathbf{R}^4$. A linear map $T : \mathbf{R}^4 \rightarrow \mathbf{R}^4$ is determined by:

$$T(u_1) = v_1, \quad T(u_2) = v_2 - v_1, \quad T(u_3) = cv_3, \quad T(u_4) = v_4 - cv_3,$$

where $c \in \mathbf{R}$ is a constant.

- (a) Determine the $(\mathcal{B}, \mathcal{C})$ -matrix of T .

[5 Marks]

- (b) For which c is T onto?

[5 Marks]

- (c) For which c is T one-to-one?

[5 Marks]

- (d) Assuming $c = 1$, give a basis for the image of T in terms of the $\mathcal{C}$ basis vectors.

[5 Marks]

- (e) Assuming $c = 1$, give a basis for the kernel of T in terms of the $\mathcal{B}$ basis vectors.

[5 Marks]

[End of Question 2: Total 25 marks]

Solution

- (a) The $(\mathcal{B}, \mathcal{C})$ -matrix of T is

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & c & -c \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- (b) T is onto when $c \neq 0$.

- (c) T is one-to-one when $c \neq 0$.

- (d) When $c = 1$, a basis for the image is (v_1, v_2, v_3, v_4) .

- (e) When $c = 1$, a basis for the kernel is $()$.

3. Subspaces, inversion, eigenvectors

Consider two subspaces U, V of $\mathbf{R}^6$. Assume that U has a basis $\mathcal{B} = (u_1, u_2)$ and that V has a basis $\mathcal{C} = (v_1, v_2, v_3, v_4)$. Assume also that the matrices

$$\begin{pmatrix} | & | & | & | & | & | \\ u_1 & u_2 & v_1 & v_2 & v_3 & v_4 \\ | & | & | & | & | & | \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

are row equivalent.

- (a) Give a basis for the subspace sum $U + V$ in terms of the vectors in the $\mathcal{B}$ and $\mathcal{C}$ bases.

[5 Marks]

- (b) Give a basis for the intersection $U \cap V$ in terms of the vectors in $\mathcal{B}$.

[5 Marks]

Consider the vector

$$x = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix} \in \mathbf{R}^3.$$

- (c) Give the reduced row echelon form of a 2×3 matrix A with $\text{Nul}(A) = \text{Span}\{x\}$.

[10 Marks]

Assume B is a 2×2 matrix that has $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ as an eigenvector with eigenvalue 1 and $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ as an eigenvector with eigenvalue -1 .

- (d) Give the inverse B^{-1} , if it exists.

[5 Marks]

[End of Question 3: Total 25 marks]

Solution

- (a) A basis for $U + V$ is $(u_1, u_2, v_1, v_2, v_4)$.

- (b) A basis for $U \cap V$ is $(u_1 + 2u_2)$.

- (c) The desired matrix is

$$A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -5 \end{pmatrix}.$$

- (d) The matrix B is its own inverse and we have

$$B^{-1} = B = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 7 & -12 \\ 4 & -7 \end{pmatrix}.$$

4. Multiple Choice Questions

Determine whether the following statements are true or false.

- (a) If A and B are invertible $n \times n$ matrices, then $I_n + A + B^{-1}$ is invertible.

[5 Marks]

- (b) If A and B are $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$.

[5 Marks]

- (c) If A and B are similar $n \times n$ matrices, then A and B have the characteristic polynomials.

[5 Marks]

- (d) If A is an $m \times n$ matrix, then $\dim(\text{Col}(A)^\perp) = \dim(\text{Row}(A)^\perp)$.

[5 Marks]

- (e) If $T : \mathbf{R}^n \rightarrow \mathbf{R}^n$ is a orthogonal projection, then $T \circ T = T$.

[5 Marks]

[End of Question 4: Total 25 marks]

Solution

- (a) False
- (b) True
- (c) True
- (d) False
- (e) True